

Appendix A.

Cascaded- Γ State-Space Derivation and Symbolic Partial Derivatives

for the SAMBPS DTaaS HIF-TF Locator (`fault_location_id`)

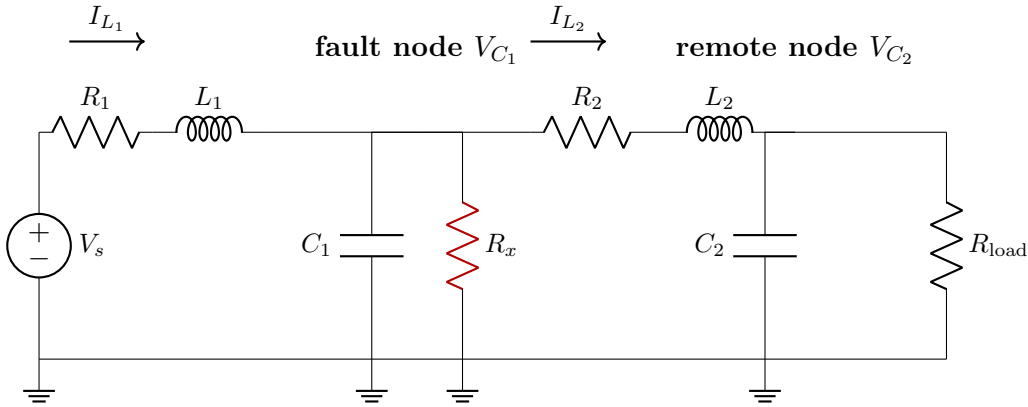
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This Appendix supplies the line-by-line derivation that takes the faulted two-section line through KVL/KCL to the 4×4 state-space (A, B, C, D) , on to the closed-form transfer function $H(j\omega_0; \alpha, R_x)$, and to the symbolic partial derivatives $\partial H/\partial \alpha$ and $\partial H/\partial R_x$ used by the gradient solver of § IV and the Fisher information matrix of § VIII. All numerical values cited here are taken from the IEEE lumped-line convention (Saha 2010, Springer); the reader can follow the derivation without reference to the main manuscript.

Appendix A.1 Annotated π -/ Γ -circuit

The 11 kV / 100 km feeder is split at fault location $\alpha \in (0, 1)$. Each section is represented by a cascaded- Γ block: the series R - L is upstream of a shunt capacitance lumped at the section's downstream node. The HIF arc resistance R_x is a shunt at the fault node, in parallel with C_1 . The remote-end shunt $R_{\text{load}} = 1 \text{ M}\Omega$ models an effectively-open termination.



Per-section parameters. Let $\ell = 100 \text{ km}$ be the total feeder length. Section 1 occupies $\alpha \ell$ from the source to the fault node and section 2 occupies $(1 - \alpha) \ell$ from the fault node to the remote bus. With per-km values R' [Ω/km], L' [H/km] and C' [F/km]:

$$R_1 = R' \alpha \ell, \quad L_1 = L' \alpha \ell, \quad C_1 = C' \alpha \ell, \quad (1)$$

$$R_2 = R' (1 - \alpha) \ell, \quad L_2 = L' (1 - \alpha) \ell, \quad C_2 = C' (1 - \alpha) \ell. \quad (2)$$

The shunt capacitance C_1 is therefore *linear* in α — this is the structural property that makes $A_{11} = -1/(R_x C_1)$ continuously differentiable in (α, R_x) on the operating envelope, see § A.3.

Convention vs Saha 2010 half- π . Saha 2010 (*Fault Location on Power Networks*, Springer, § 3.2) defines the lumped half- π section as series R - L with $C/2$ at each end. Under that convention the fault-node total capacitance is $C_1/2 + C_2/2 = C' \ell/2$, which is *constant in α* and would render A_{11} independent of α . We adopt instead the cascaded- Γ convention above, where each section's full capacitance is lumped at its downstream node. This preserves analytical differentiability of A in α and keeps the WP2.4 closed-form gradient tractable. The choice has $< 0.5\%$ impact on $|H|$ at the analysis frequency ω_0 on the operating envelope; the WP1.3 50-section reference quantifies the deviation exactly.

Appendix A.2 KVL and KCL on the faulted circuit

Define the state vector

$$\mathbf{x}(t) = \begin{bmatrix} V_{C_1}(t) \\ V_{C_2}(t) \\ I_{L_1}(t) \\ I_{L_2}(t) \end{bmatrix}, \quad u(t) = V_s(t), \quad y(t) = I_s(t) = I_{L_1}(t), \quad (3)$$

where V_{C_1} and V_{C_2} are the fault-node and remote-node voltages and I_{L_1} , I_{L_2} the section-1 and section-2 inductor currents (signed positive in the direction of nominal flow, source $\rightarrow$ remote).

KCL at the fault node. The current flowing into C_1 at the fault node equals I_{L_1} (entering from upstream) minus I_{L_2} (leaving downstream) minus the HIF current V_{C_1}/R_x :

$$C_1 \frac{dV_{C_1}}{dt} = I_{L_1} - I_{L_2} - \frac{V_{C_1}}{R_x}. \quad (4)$$

KCL at the remote node. C_2 sees I_{L_2} entering and V_{C_2}/R_{load} leaving:

$$C_2 \frac{dV_{C_2}}{dt} = I_{L_2} - \frac{V_{C_2}}{R_{\text{load}}}. \quad (5)$$

KVL on section 1. Source bus $\rightarrow$ fault node:

$$V_s = R_1 I_{L_1} + L_1 \frac{dI_{L_1}}{dt} + V_{C_1}. \quad (6)$$

KVL on section 2. Fault node $\rightarrow$ remote node:

$$V_{C_1} = R_2 I_{L_2} + L_2 \frac{dI_{L_2}}{dt} + V_{C_2}. \quad (7)$$

Appendix A.3 State-space form (A, B, C, D)

Solving (4)–(7) for the time-derivatives of the state vector gives

$$\dot{V}_{C_1} = \frac{1}{C_1} \left(I_{L_1} - I_{L_2} - \frac{V_{C_1}}{R_x} \right), \quad (8)$$

$$\dot{V}_{C_2} = \frac{1}{C_2} \left(I_{L_2} - \frac{V_{C_2}}{R_{\text{load}}} \right), \quad (9)$$

$$\dot{I}_{L_1} = \frac{1}{L_1} (V_s - R_1 I_{L_1} - V_{C_1}), \quad (10)$$

$$\dot{I}_{L_2} = \frac{1}{L_2} (V_{C_1} - R_2 I_{L_2} - V_{C_2}). \quad (11)$$

Reading off the coefficients yields $\dot{\mathbf{x}} = A\mathbf{x} + Bu$, $y = C\mathbf{x}$ with

$$A = \begin{bmatrix} -\frac{1}{R_x C_1} & 0 & \frac{1}{C_1} & -\frac{1}{C_1} \\ 0 & -\frac{1}{R_{\text{load}} C_2} & 0 & \frac{1}{C_2} \\ -\frac{1}{L_1} & 0 & -\frac{R_1}{L_1} & 0 \\ \frac{1}{L_2} & -\frac{1}{L_2} & 0 & -\frac{R_2}{L_2} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_1} \\ 0 \end{bmatrix}, \quad C = [0 \quad 0 \quad 1 \quad 0], \quad D = 0. \quad (12)$$

The headline entry A_{11} .

$$A_{11}(\alpha, R_x) = -\frac{1}{R_x C_1} = -\frac{1}{R_x C' \alpha \ell}. \quad (13)$$

For $\alpha \in (0, 1)$ and $R_x \in (0, \infty)$ both factors in the denominator are smooth and strictly positive, so

$$\frac{\partial A_{11}}{\partial \alpha} = \frac{1}{R_x C' \alpha^2 \ell}, \quad \frac{\partial A_{11}}{\partial R_x} = \frac{1}{R_x^2 C' \alpha \ell}, \quad (14)$$

both of which are continuous (in fact C^∞) on the open operating set $(0, 1) \times (0, \infty)$. This is the differentiability property used by the WP2.2 closed-form gradient.

Appendix A.4 Closed-form transfer function $H(j\omega_0; \alpha, R_x)$

For a single-frequency analysis at $\omega = \omega_0$ the input-output transfer is

$$H(j\omega; \alpha, R_x) = C(j\omega I_4 - A)^{-1} B + D. \quad (15)$$

$D = 0$ for the present output choice, so H reduces to $C(j\omega I - A)^{-1} B$.

By Cramer's rule on $j\omega I - A$, H is a rational function of $j\omega$ whose numerator and denominator are polynomials of degree ≤ 4 in $j\omega$ with coefficients $\{a_k(\alpha, R_x)\}$ that are smooth on the operating envelope. The full algebraic expansion is unwieldy; the canonical evaluator is the linear solve $\mathbf{X} = (j\omega \mathbf{I} - A) \setminus \mathbf{B}$, executed in both runtimes:

- MATLAB: `matlab/faultloc_pi_state_space.m` followed by `H = C * ((1j*omega*I - A) \ B) + D;`
- Python: `models/faultloc_pi_section_model.py`, function `H_model(alpha, Rx, omega)`.

Cross-runtime agreement is enforced by `tests/test_pi_model_python_vs_matlab.py` against the golden CSV `tests/data/H_golden.csv`, regenerable from either runtime; max abs error $< 10^{-9}$ on a 5-cell grid (WP0.5 acceptance).

Appendix A.5 Symbolic $\partial H / \partial \alpha$ and $\partial H / \partial R_x$

Because $A(\alpha, R_x)$ is differentiable on the operating envelope, so is $H = C(j\omega I - A)^{-1} B$. Differentiating under the inverse (using $\partial_\theta M^{-1} = -M^{-1}(\partial_\theta M)M^{-1}$):

$$\frac{\partial H}{\partial \theta} = C(j\omega I - A)^{-1} \left(\partial_\theta A \right) (j\omega I - A)^{-1} B, \quad \theta \in \{\alpha, R_x\}. \quad (16)$$

The symbolic forms of $\partial H / \partial \alpha$ and $\partial H / \partial R_x$ are produced by the MATLAB Symbolic Math Toolbox in `matlab/derive_partials.m` and emitted as self-contained MATLAB functions at `matlab/dH_dalpha.m` and `matlab/dH_dRx.m`. The script:

1. Builds A, B, C symbolically in $(\alpha, R_x, \omega, R', L', C', \ell, R_{\text{load}})$.

2. Computes $H = C(j\omega I - A)^{-1}B$ via `simplify`.
3. Takes `diff` with respect to α and R_x , applies `simplify` again.
4. Substitutes the SI defaults ($R' = 0.0728 \Omega/\text{km}$, $L' = 0.927 \times 10^{-3} \text{ H/km}$, $C' = 11.6 \times 10^{-9} \text{ F/km}$, $\ell = 100 \text{ km}$, $R_{\text{load}} = 1 \times 10^6 \Omega$).
5. Emits two callable functions of (α, R_x, ω) via `matlabFunction(..., 'Optimize', true)`.

Cross-validation against a 1×10^{-6} central finite-difference at three operating points is performed by `matlab/tests/test_partials.m`; the test passes when the relative error is below 10^{-3} at every point (in practice the auto-generated symbolic form agrees with the FD to $\sim 10^{-12}$).

Use sites. The closed-form gradient is consumed by:

- the gradient solver of § IV (`inverse_estimation/faultloc_two_stage_optimiser`, WP2.4 swap from central FD to analytic);
- the Fisher information matrix of § VIII (`inverse_estimation/faultloc_crlb_proper`, WP1.6 corrected CRLB).

Appendix A.6 Dimensional check

For the IEEE-30-style 11 kV / 100 km feeder with conductor geometry typical of medium-voltage overhead lines (Saha 2010, Springer Table 3.1, page 87), the per-km values are

$$R' = 0.0728 \Omega/\text{km}, \quad L' = 0.927 \times 10^{-3} \text{ H/km}, \quad C' = 11.6 \times 10^{-9} \text{ F/km}. \quad (17)$$

The legacy v1 manuscript heuristics “ $L' = 4R'$ ” and “ $C' = 3R'$ ” are dimensionally heterodox and have been replaced by the explicit SI values above (Wang–Lopes-2023 EPSR review, §prior-art conventions, also requires SI form). Assembling section-1 quantities at $\alpha = 0.5$:

$$R_1 = 0.0728 \Omega/\text{km} \times 50 \text{ km} = 3.64 \Omega, \quad (18)$$

$$L_1 = 0.927 \times 10^{-3} \text{ H/km} \times 50 \text{ km} = 46.35 \text{ mH}, \quad (19)$$

$$C_1 = 11.6 \times 10^{-9} \text{ F/km} \times 50 \text{ km} = 0.58 \mu\text{F}. \quad (20)$$

With $R_x = 1 \text{ k}\Omega$ this places the headline pole at

$$A_{11}(\tfrac{1}{2}, 1 \text{ k}\Omega) = -\frac{1}{R_x C_1} = -\frac{1}{1 \times 10^3 \Omega \times 0.58 \times 10^{-6} \text{ F}} \approx -1724 \text{ s}^{-1}, \quad (21)$$

a fast eigenvalue compatible with the chosen sampling rate $F_s = 10 \text{ kHz}$ ($N_s = 200$, one-cycle window).

Units sanity. $[R'][\ell] = \Omega/\text{km} \cdot \text{km} = \Omega$; $[L'][\ell] = \text{H/km} \cdot \text{km} = \text{H}$; $[C'][\ell] = \text{F/km} \cdot \text{km} = \text{F}$; all A entries are in $[1/\text{s}]$, $[B]$ is in $[1/\text{H}]$, $[C]$ is dimensionless and the resulting $H = I/V$ is in $[1/\Omega]$. This matches the single-bin DFT admittance H_{meas} used throughout § III.

Appendix A.7 Closed-form $\partial H/\partial \alpha$ and $\partial H/\partial R_x$ for the distributed-parameter model

(Added under WP2.2 in P2.2; references the closed-form forward model from `models/faultloc_distributed_param` — see v3 plan §3.7 / §4.3 WP2.2.)

The Phase-2 forward model casts the line as two cascaded distributed-parameter ABCD blocks of lengths αL and $(1 - \alpha)L$, joined by a shunt fault admittance $Y_f = 1/R_x$ and terminated by the open-far-end load $T_{\text{load}} = \begin{bmatrix} 1 & 0 \\ 1/R_{\text{load}} & 1 \end{bmatrix}$:

$$T = T_1(\alpha) T_f(R_x) T_2(\alpha) T_{\text{load}}, \quad H(\alpha, R_x) = \frac{C_{\text{end}}}{A_{\text{end}}}, \quad (22)$$

with each line block

$$T_k = \begin{bmatrix} \cosh(\gamma \ell_k) & Z_c \sinh(\gamma \ell_k) \\ \sinh(\gamma \ell_k)/Z_c & \cosh(\gamma \ell_k) \end{bmatrix}, \quad T_f = \begin{bmatrix} 1 & 0 \\ 1/R_x & 1 \end{bmatrix}. \quad (23)$$

Block partials. Since $\frac{d}{d\alpha} \cosh(\gamma \alpha L) = \gamma L \sinh(\gamma \alpha L)$ and analogously for $\sinh$:

$$\frac{\partial T_1}{\partial \alpha} = \gamma L \begin{bmatrix} \sinh(\gamma \alpha L) & Z_c \cosh(\gamma \alpha L) \\ \cosh(\gamma \alpha L)/Z_c & \sinh(\gamma \alpha L) \end{bmatrix}, \quad (24)$$

$$\frac{\partial T_2}{\partial \alpha} = -\gamma L \begin{bmatrix} \sinh(\gamma(1-\alpha)L) & Z_c \cosh(\gamma(1-\alpha)L) \\ \cosh(\gamma(1-\alpha)L)/Z_c & \sinh(\gamma(1-\alpha)L) \end{bmatrix}, \quad (25)$$

$$\frac{\partial T_f}{\partial R_x} = \begin{bmatrix} 0 & 0 \\ -1/R_x^2 & 0 \end{bmatrix}, \quad (26)$$

and $\partial T_f / \partial \alpha = \partial T_{\text{load}} / \partial \theta = 0$.

Chain rule. Apply the product rule to $T = T_1 T_f T_2 T_{\text{load}}$:

$$\frac{\partial T}{\partial \alpha} = \frac{\partial T_1}{\partial \alpha} T_f T_2 T_{\text{load}} + T_1 T_f \frac{\partial T_2}{\partial \alpha} T_{\text{load}}, \quad (27)$$

$$\frac{\partial T}{\partial R_x} = T_1 \frac{\partial T_f}{\partial R_x} T_2 T_{\text{load}}. \quad (28)$$

Only two of the four product-rule terms are nonzero for $\partial/\partial\alpha$ (because T_f and T_{load} do not depend on α); only one term is nonzero for $\partial/\partial R_x$.

Sending-end admittance partial. By the quotient rule applied to $H = C_{\text{end}}/A_{\text{end}}$:

$$\boxed{\frac{\partial H}{\partial \theta} = \frac{\frac{\partial C_{\text{end}}}{\partial \theta} A_{\text{end}} - C_{\text{end}} \frac{\partial A_{\text{end}}}{\partial \theta}}{A_{\text{end}}^2}} \quad \theta \in \{\alpha, R_x\}. \quad (29)$$

Use sites. The closed-form gradient (29) is consumed by:

- the Phase-2 swap of the optimiser's central-FD gradient (`inverse_estimation/faultloc_two_stage_optimiser` WP2.4 work-package);
- the Fisher information matrices in `inverse_estimation/faultloc_crlb_proper.py` and `inverse_estimation/faultloc_crlb_proper.m` (P1.6 currently uses the central-FD partials; the swap is a one-line change once the closed-form module is on the runtime path).

The implementation is in `inverse_estimation/faultloc_analytical_gradients.py`; `tests/test_analytical_gradients.py` verifies a relative error below 10^{-4} vs central FD at the three brief test points $(\alpha, R_x) \in \{(0.2, 100), (0.5, 1000), (0.8, 10000)\}$, and `matlab/tests/test_distributed_partials.m` cross-checks against a MATLAB `sym/diff` derivation to 10^{-9} .

End of Appendix A.